

Step 1: Software Requirements and Project Scope

Project Title:

Local Service Problem Diagnosis and Matching System

1. Problem Definition and Project Scope

Many local users have difficulty finding the right service provider when a household appliance or piece of equipment stops working. A user may know what is wrong in general, but may not know which type of service provider can handle the problem. This can lead to contacting several providers before finding the right one.

The proposed system will give local users one place to submit their service needs and find suitable providers. Users who already know the required service can select the service type directly. Users who are unsure can describe the problem in their own words. The NLP/ML component will analyze the description and suggest a suitable service category. The system will then use that category to help find matching service providers. The main users are local users, service providers, and the system administrator. The NLP/ML component will support the system by classifying problem descriptions. The system will store user accounts, provider information, service requests, service categories, request status, and service history in SQLite.

The project will include user and provider registration, problem submission, service-category classification, provider matching, service-request tracking, provider status updates, feedback, and administrator management. The project will not include online payments, GPS tracking, emergency dispatch, mobile applications, or automatic technical diagnosis of equipment problems. The main purpose is to make it easier for local users to identify the appropriate service category and connect with a suitable local service provider